

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A
(Objective Type Questions)
(Compulsory)

1. Choose the correct answer of the following :

2×10 = 20

- (a) The null hypothesis is denoted by :

(i) H_0

(ii) H_1

(iii) H

(iv) H_2

- (b) The variance of χ^2 with n degree of freedom is :

(i) $2n$

(ii) n

(iii) n^2

(iv) $3n$

(c) The degree of freedom of error of R. B. D. is :

- (i) $(r-1)(t-1)$
- (ii) $(t-1)$
- (iii) $(r-1)$
- (iv) $(r-1)(t-1)$ same (i)

(d) For testing $H_0 : \mu_1 = \mu_2$, the suitable test statistic is :

- (i) t
- (ii) χ^2
- (iii) F
- (iv) None of these

(e) The sample mean is _____ estimate of the population mean.

- (i) Biased
- (ii) Positive Biased
- (iii) Unbiased
- (iv) None of these

(f) The M. G. F. of χ^2 distribution is :

- (i) $(1-2t)^{-n}$
- (ii) $(1-2t)^{-2n}$
- (iii) $(1-2t)^{-n/2}$
- (iv) $(1-t)$

- (g) Rejection of correct hypothesis is :
- (i) Type II error
 - (ii) Sampling error
 - (iii) Type I error
 - (iv) None of these
- (h) There are _____ basic principles of design.
- (i) 2
 - (ii) 1
 - (iii) 3
 - (iv) 4
- (i) The range of t variate is :
- (i) 0 to ∞
 - (ii) $-\infty$ to 0
 - (iii) $-\infty$ to ∞
 - (iv) 0 to 1
- (j) The test is called one tail or two tail test which depend on :
- (i) Null hypothesis
 - (ii) Level of significance
 - (iii) Alternative hypotheses
 - (iv) None of these

Group – B

(Short-answer Type Questions)

Answer any **four** questions of the following :

$5 \times 4 = 20$

2. Define Completely Randomised Design (CRD).

ES – 7/2

(3)

(Turn over)

3. Obtain mean of χ^2 distribution.
4. Define Null and Alternative hypotheses.
5. Discuss student's t distribution.
6. Show that sample mean is unbiased estimate of the population mean.
7. Name the test statistic for testing $H_0 : \mu_1 = \mu_2$, and also define the procedures for testing.

Group – C

(Long-answer Type Questions)

Answer any **three** questions of the following :

$$10 \times 3 = 30$$

8. What are the criteria of a good estimate ? Explain each with example.
9. Discuss χ^2 distribution. Find its Variance and M. G. F.
10. Discuss the three basic principles of design.
11. Obtain the M. L. E of the unknown parameter (p) of Binomial distribution when n is known.
12. What is Randomised Block Design ? Give its layout and analysis of variance.

ES – 7/2 (500,

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(4)

Voc(Sem-IV) — BCA
(GE – 4) Stat – II